

THE UNIFORM $\mathbb{G}_0$ DICHOTOMY

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ABSTRACT. We give a classical proof of the generalization of the uniform $\mathbb{G}_0$ dichotomy to families of digraphs indexed by elements of $\kappa^{\mathbb{N}}$, and obtain Kanovei's generalization of the $\mathbb{G}_0$ dichotomy to κ -Souslin digraphs as a corollary.

INTRODUCTION

Suppose that κ is an infinite cardinal and X is a Hausdorff space. Endow κ with the discrete topology and $\kappa^{\mathbb{N}}$ with the product topology. For all $n \in \mathbb{N}$ and $s \in \kappa^n$, set $|s| = n$. For all $\sigma \in \kappa^{\leq \mathbb{N}}$, let $s \sqsubseteq \sigma$ indicate that $\forall i < |s| \ s(i) = \sigma(i)$. Define $\mathcal{N}_s = \{\sigma \in \kappa^{\mathbb{N}} \mid s \sqsubseteq \sigma\}$. We write $\forall^* \sigma \in \kappa^{\mathbb{N}} \ \phi(\sigma)$ to indicate that $\{\sigma \in \kappa^{\mathbb{N}} \mid \phi(\sigma)\}$ is comeager. The *projection* function from $\prod_{i \in I} X_i$ to X_i is given by $\text{proj}_{X_i}(x) = x(i)$. The *horizontal and vertical sections* of a set $R \subseteq X \times Y$ are given by $R^y = \{x \in X \mid x R y\}$ and $R_x = \{y \in Y \mid x R y\}$. Given a function $\pi: R \subseteq X \times Y \rightarrow Z$, define $\pi_x: R_x \rightarrow Z$ by $\pi_x(y) = \pi(x, y)$.

We say that $\phi: C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \rightarrow \kappa^{\mathbb{N}} \times X$ is *parameter preserving* if $\text{proj}_{\kappa^{\mathbb{N}}} \circ \phi = \text{proj}_{\kappa^{\mathbb{N}}} \upharpoonright C$, and $A \subseteq \kappa^{\mathbb{N}} \times X$ has *uniformly analytic (vertical) sections* if there are a closed set $C \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and a continuous parameter-preserving function $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ such that $A = \phi(C)$.

The *diagonal union* of a sequence $(X_s)_{s \in \kappa^{< \mathbb{N}}}$ of subsets of $\kappa^{\mathbb{N}} \times X$ is given by $\bigvee_{s \in \kappa^{< \mathbb{N}}} X_s = \bigcup_{s \in \kappa^{< \mathbb{N}}} X_s \cap (\mathcal{N}_s \times X)$. Countable unions are expressible as diagonal unions, and the vertical sections of diagonal unions are themselves countable unions. We say that a set $B \subseteq \kappa^{\mathbb{N}} \times X$ has *uniformly Borel (vertical) sections* if it is in the smallest family of subsets of $\kappa^{\mathbb{N}} \times X$ containing the products of $\kappa^{\mathbb{N}}$ with open subsets of X and closed under complements and diagonal unions. We say that a function $c: \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ is *weakly Borel* if $c^{-1}(\{n\})$ has uniformly Borel sections for all $n \in \mathbb{N}$.

A digraph on X is an irreflexive set $G \subseteq X \times X$. A *homomorphism* from a digraph G on X to a digraph H on Y is a function $\pi: X \rightarrow Y$ such that $(\pi \times \pi)(G) \subseteq H$. A *Y -coloring* of G is a homomorphism $c: X \rightarrow Y$ from G to inequality. Fix $\mathfrak{s} \in \prod_{n \in \mathbb{N}} 2^n$ such that $\mathfrak{s}(\mathbb{N})$ is dense in $2^{< \mathbb{N}}$ and define $\mathbb{G}_0 = \bigcup_{n \in \mathbb{N}} \{(\mathfrak{s}_n \frown (i) \frown c)_{i < 2} \mid c \in 2^{\mathbb{N}}\}$.

The uniform $\mathbb{G}_0$ dichotomy for digraphs indexed by $\mathbb{N}^{\mathbb{N}}$ is a straightforward corollary of the $\mathbb{G}_0$ dichotomy itself (see [Mil12, Theorem 31]). Here we show that the following generalization to digraphs indexed by $\kappa^{\mathbb{N}}$ is a consequence of the classical proof of the $\mathbb{G}_0$ dichotomy:

Theorem 1. *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ has uniformly analytic sections that are digraphs on X . Then exactly one of the following holds:*

- (1) *There is a weakly Borel function $c: \kappa^{\mathbb{N}} \times X \rightarrow \mathbb{N}$ such that c_σ is a coloring of G_σ for all $\sigma \in \kappa^{\mathbb{N}}$.*
- (2) *There is a continuous homomorphism $\pi: 2^{\mathbb{N}} \rightarrow X$ from $\mathbb{G}_0$ to G_σ for some $\sigma \in \kappa^{\mathbb{N}}$.*

In §1, we note closure properties of sets with uniformly analytic and Borel sections. In §2, we consider how these notions interact with Baire category. In §3, we establish a uniform version of Lusin's separation theorem. In §4, we prove Theorem 1. And in §5, we obtain Kanovei's κ -Souslin generalization [Kan97, Theorem 2] of the $\mathbb{G}_0$ dichotomy.

We leave it to the reader to check that these results can be combined with the arguments establishing the usual consequences of the $\mathbb{G}_0$ dichotomy to obtain analogous uniform and κ -Souslin generalizations.

1. POINTCLASSES

We direct the reader to [Kec95, §14] for the most basic facts concerning analytic sets. The following closure properties can be proven in the same fashion as the corresponding facts for analytic sets:

Proposition 1.1. *Suppose that κ is an aleph and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under countable intersections and countable unions.*

Proposition 1.2. *Suppose that κ is an aleph, X and Y are Hausdorff spaces, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $B \subseteq \kappa^{\mathbb{N}} \times Y$ have uniformly analytic sections, and $f: A \rightarrow \kappa^{\mathbb{N}} \times Y$ is continuous and parameter preserving. Then $f(A)$ and $f^{-1}(B)$ have uniformly analytic sections.*

Given a set $S \subseteq \kappa^{<\mathbb{N}}$ and a sequence $(X_s)_{s \in S}$ of subsets of $\kappa^{\mathbb{N}} \times X$, define $\bigvee_{s \in S} X_s = \bigvee_{s \in \kappa^{<\mathbb{N}}} X_s$, where $X_s = \emptyset$ for all $s \in \sim S$.

Proposition 1.3. *Suppose that κ is an aleph and X is a Hausdorff space. Then the subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections are closed under diagonal unions.*

Proof. Suppose that $(A_s)_{s \in \kappa^{<\mathbb{N}}}$ is a sequence of subsets of $\kappa^{\mathbb{N}} \times X$ with uniformly analytic sections and $A = \bigvee_{s \in \kappa^{<\mathbb{N}}} A_s$. Then there are closed

sets $C_s \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi_s: C_s \rightarrow \kappa^{\mathbb{N}} \times X$ with $A_s = \phi_s(C_s)$ for all $s \in \kappa^{<\mathbb{N}}$. As the set $C_n = \bigcap_{s \in \kappa^n} C_s$ is closed and the function $\phi_n = \bigcup_{s \in \kappa^n} \phi_s \upharpoonright (C_s \cap (\mathcal{N}_s \times \mathbb{N}^{\mathbb{N}}))$ is continuous, the set $A_n = \bigcap_{s \in \kappa^n} A_s = \phi_n(C_n)$ has uniformly analytic sections for all $n \in \mathbb{N}$, as does $A = \bigcup_{n \in \mathbb{N}} A_n$ by Proposition 1.1. $\square$

Proposition 1.4. *Suppose that κ is an aleph and X is a topological space. Then every Borel subset of $\kappa^{\mathbb{N}} \times X$ has uniformly Borel sections.*

Proof. It is sufficient to show that every open set $U \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. For all $s \in \kappa^{<\mathbb{N}}$, let $U_s \subseteq X$ be the maximal open set for which $\mathcal{N}_s \times U_s \subseteq U$. Then $U = \bigcap_{s \in \kappa^{<\mathbb{N}}} \kappa^{\mathbb{N}} \times U_s$. $\square$

Proposition 1.5. *Suppose that κ is an aleph, X and Y are topological spaces, $B \subseteq \kappa^{\mathbb{N}} \times Y$ has uniformly Borel sections, and $f: \kappa^{\mathbb{N}} \times X \rightarrow \kappa^{\mathbb{N}} \times Y$ is Borel and parameter preserving. Then $f^{-1}(B)$ has uniformly Borel sections.*

Proof. The family of sets for which the proposition holds contains the open sets by Proposition 1.4 and is clearly closed under complements. The fact that f is parameter preserving ensures that it is also closed under diagonal unions. $\square$

Proposition 1.6. *Suppose that X is a Hausdorff space, $A \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly analytic sections, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then $A \cap B$ and $A \setminus B$ have uniformly analytic sections.*

Proof. By the usual proof of the corresponding fact for analytic and Borel sets, we need only note that if $B = \bigcap_{s \in \kappa^{<\mathbb{N}}} B_s$ and both $A \cap B_s$ and $A \setminus B_s$ have uniformly analytic sections for all $s \in \kappa^{<\mathbb{N}}$, then $A \cap B = \bigcap_{s \in \kappa^{<\mathbb{N}}} A \cap B_s$ has uniformly analytic sections by Proposition 1.3, and $A \setminus B = \bigcap_{n \in \mathbb{N}} \bigcap_{s \in \kappa^n} A \setminus B_s$ has uniformly analytic sections by Propositions 1.1 and 1.3. $\square$

2. BAIRE CATEGORY

We direct the reader to [Kec95, §8] for the most basic facts concerning Baire category (such as localization). Although $\kappa^{\mathbb{N}}$ need not be separable, the usual proofs use only the Baire category theorem and go through unchanged for completely metrizable spaces.

Proposition 2.1. *Suppose that κ is an aleph, X is a topological space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then every horizontal section of B has the Baire property.*

Proof. It is enough to show that if $n \in \mathbb{N}$ and the horizontal sections of $B_s \subseteq \kappa^{\mathbb{N}} \times X$ have the Baire property for all $s \in \kappa^{<\mathbb{N}}$, then the horizontal

sections of $B = \bigcap_{s \in \kappa^n} B_s$ also have the Baire property. Given $x \in X$, fix an open set $U_s \subseteq \kappa^\mathbb{N}$ and dense open sets $U_{m,s} \subseteq \kappa^\mathbb{N}$ such that $(B_s^x \triangle U_s) \cap \bigcap_{m \in \mathbb{N}} U_{m,s} = \emptyset$ for all $s \in \kappa^n$. Then $U = \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap U_s$ is open, $U_m = \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap U_{m,s}$ is dense and open for all $m \in \mathbb{N}$, and $(B^x \triangle U) \cap \bigcap_{m \in \mathbb{N}} U_m = \emptyset$, so B^x has the Baire property. $\square$

3. SEPARATION

Recall that $B \subseteq X$ separates $A_0, A_1 \subseteq X$ if $A_0 \subseteq B$ and $A_1 \cap B = \emptyset$.

Proposition 3.1. *Suppose that κ is an aleph, X is a Hausdorff space, and $A_0, A_1 \subseteq \kappa^\mathbb{N} \times X$ are disjoint sets with uniformly analytic sections. Then they are separated by a set with uniformly Borel sections.*

Proof. Fix closed sets $C_i \subseteq \kappa^\mathbb{N} \times \mathbb{N}^\mathbb{N}$ and continuous parameter-preserving functions $\phi_i: C_i \rightarrow \kappa^\mathbb{N} \times X$ such that $A_i = \phi_i(C_i)$, and define $A_{i,s,t} = \phi_i(C_i \cap (\mathcal{N}_s \times \mathcal{N}_t))$ for all $i < 2$, $s \in \kappa^{<\mathbb{N}}$, and $t \in \mathbb{N}^{<\mathbb{N}}$.

Following the usual convention, we identify $\bigcup_{n \in \mathbb{N}} \kappa^n \times \mathbb{N}^n \times \mathbb{N}^n$ with $(\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$. Set $T = \{(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}} \mid A_{0,s,t} \cap \overline{A_{1,s,u}} \neq \emptyset\}$.

Lemma 3.2. *The tree T is well-founded.*

Proof. Suppose, towards a contradiction, that $(\sigma, c_0, c_1) \in [T]$. Then $(\sigma, c_i) \in C_i$ for all $i < 2$. Set $x_i = (\text{proj}_X \circ \phi_i)(\sigma, c_i)$ for all $i < 2$. As A_0 and A_1 are disjoint, it follows that $x_0 \neq x_1$, so there are disjoint open neighborhoods U_0 and U_1 of x_0 and x_1 . Fix $n \in \mathbb{N}$ such that $A_{i,\sigma \upharpoonright n, c_i \upharpoonright n} \subseteq \kappa^\mathbb{N} \times U_i$ for all $i < 2$. Then $A_{0,\sigma \upharpoonright n, c_0 \upharpoonright n} \cap \overline{A_{1,\sigma \upharpoonright n, c_1 \upharpoonright n}} = \emptyset$, which contradicts the fact that $(\sigma \upharpoonright n, c_0 \upharpoonright n, c_1 \upharpoonright n) \in T$. $\square$

Recursively define $B_{s,t,u} \subseteq \kappa^\mathbb{N} \times X$ for $(s, t, u) \in (\kappa \times \mathbb{N} \times \mathbb{N})^{<\mathbb{N}}$ by

$$B_{s,t,u} = \begin{cases} \overline{\bigcap_{r \in S_s} \bigcup_{m \in \mathbb{N}} \bigcap_{n \in \mathbb{N}} B_{r,t \hat{\smallfrown} (m), u \hat{\smallfrown} (n)}} & \text{if } (s, t, u) \in T \text{ and} \\ \sim A_{1,s,u} & \text{otherwise,} \end{cases}$$

where $S_s = \{s \hat{\smallfrown} (\alpha) \mid \alpha < \kappa\}$. The obvious induction shows that $B_{s,t,u}$ has uniformly Borel sections and separates $A_{0,s,t}$ from $A_{1,s,u}$, so $B_{\emptyset, \emptyset, \emptyset}$ separates A_0 from A_1 . $\square$

Given sets $A_i \subseteq \kappa^\mathbb{N} \times X_i$ for all $i < 2$ and $R \subseteq \kappa^\mathbb{N} \times (X_0 \times X_1)$, we say that (A_0, A_1) is R -independent if $\forall \sigma \in \kappa^\mathbb{N} ((A_0)_\sigma \times (A_1)_\sigma) \cap R_\sigma = \emptyset$.

Proposition 3.3. *Suppose that κ is an aleph, X_0, X_1 are Hausdorff spaces, $A_i \subseteq \kappa^\mathbb{N} \times X_i$ for all $i < 2$ and $R \subseteq \kappa^\mathbb{N} \times (X_0 \times X_1)$ have uniformly analytic sections, and (A_0, A_1) is R -independent. Then there are sets $B_i \subseteq \kappa^\mathbb{N} \times X_i$ for all $i < 2$ with uniformly Borel sections such that $A_0 \subseteq B_0$, $A_1 \subseteq B_1$, and (B_0, B_1) is R -independent.*

Proof. The set $A'_0 = \text{proj}_{\kappa^{\mathbb{N}} \times X_0}((\text{proj}_{\kappa^{\mathbb{N}} \times X_1} \upharpoonright R)^{-1}(A_1))$ has uniformly analytic sections by two applications of Proposition 1.2. As $A_0 \cap A'_0 = \emptyset$, Proposition 3.1 yields a set $B_0 \subseteq \kappa^{\mathbb{N}} \times X_0$ with uniformly Borel sections separating A_0 from A'_0 .

Along similar lines, the set $A'_1 = \text{proj}_{\kappa^{\mathbb{N}} \times X_1}((\text{proj}_{\kappa^{\mathbb{N}} \times X_0} \upharpoonright R)^{-1}(B_0))$ has uniformly analytic sections by Propositions 1.2, 1.5, and 1.6. As $A_1 \cap A'_1 = \emptyset$, Proposition 3.1 yields a set $B_1 \subseteq \kappa^{\mathbb{N}} \times X_1$ with uniformly Borel sections separating A_1 from A'_1 . $\square$

Given a set $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$, we say that A is *G-independent* if (A, A) is *G-independent*.

Proposition 3.4. *Suppose that κ is an aleph, X is a Hausdorff space, $A \subseteq \kappa^{\mathbb{N}} \times X$ and $G \subseteq \kappa^{\mathbb{N}} \times (X \times X)$ have uniformly analytic sections, and A is *G-independent*. Then there exists $B \subseteq \kappa^{\mathbb{N}} \times X$ with uniformly Borel sections such that $A \subseteq B$ and B is *G-independent*.*

Proof. By Proposition 3.3, there exist $B_0, B_1 \subseteq \kappa^{\mathbb{N}} \times X$ with uniformly Borel sections such that $A \subseteq B_0 \cap B_1$ and (B_0, B_1) is *G-independent*. Then the set $B_0 \cap B_1$ is as desired. $\square$

4. THE UNIFORM $\mathbb{G}_0$ DICHOTOMY

We use the notation and basic facts concerning $\mathbb{G}_0$, colorings, and homomorphisms from [dRM21, §1].

Proof of Theorem 1. Define $p_i: \kappa^{\mathbb{N}} \times (X \times X) \rightarrow \kappa^{\mathbb{N}} \times X$ by setting $p_i(\sigma, (x_0, x_1)) = (\sigma, x_i)$ for all $i < 2$, $\sigma \in \kappa^{\mathbb{N}}$, and $x_0, x_1 \in X$. As Propositions 1.1 and 1.2 ensure that $\bigcup_{i < 2} p_i(G)$ has uniformly analytic sections, there are closed sets $C, D \subseteq \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}}$ and continuous parameter-preserving functions $\phi: C \rightarrow \kappa^{\mathbb{N}} \times X$ and $\psi: D \rightarrow \kappa^{\mathbb{N}} \times (X \times X)$ such that $\bigcup_{i < 2} p_i(G) = \phi(C)$ and $G = \psi(D)$.

An *approximation* is a triple $a = (n^a, \phi^a, (\psi_k^a)_{k < n^a})$, where $n^a \in \mathbb{N}$, $\phi^a: 2^{n^a} \rightarrow \mathbb{N}^{n^a}$, and $\psi_k^a: 2^{n^a - (k+1)} \rightarrow \mathbb{N}^{n^a}$ for all $k < n^a$. An approximation b is a *one-step extension* of a if:

- (1) $n^b = n^a + 1$,
- (2) $\forall u \in 2^{n^a} \forall v \in 2^{n^b} (u \sqsubseteq v \implies \phi^a(u) \sqsubseteq \phi^b(v))$, and
- (3) $\forall k < n^a \forall u \in 2^{n^a - (k+1)} \forall v \in 2^{n^b - (k+1)} (u \sqsubseteq v \implies \psi_k^a(u) \sqsubseteq \psi_k^b(v))$.

Given $\sigma \in \kappa^{\mathbb{N}}$, a σ -*configuration* is a triple $\gamma = (n^\gamma, \phi^\gamma, (\psi_k^\gamma)_{k < n^\gamma})$, where $\phi^\gamma: 2^{n^\gamma} \rightarrow \mathbb{N}^{n^\gamma}$ and $\psi_k^\gamma: 2^{n^\gamma - (k+1)} \rightarrow \mathbb{N}^{n^\gamma}$ for all $k < n^\gamma$, such that:

- (1) $\forall u \in 2^{n^\gamma} \phi^\gamma(u) \in C_\sigma$,
- (2) $\forall k < n^\gamma \forall v \in 2^{n^\gamma - (k+1)} \psi_k^\gamma(v) \in D_\sigma$, and
- (3) $\forall k < n^\gamma \forall v \in 2^{n^\gamma - (k+1)}$
 $(\text{proj}_{X \times X} \circ \psi_\sigma \circ \psi_k^\gamma)(v) = ((\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(s_k \frown (i) \frown v))_{i < 2}$.

We say that γ is *compatible* with $Y \subseteq X$ if $(\text{proj}_X \circ \phi_\sigma \circ \phi^\gamma)(2^{n^\gamma}) \subseteq Y$, and *compatible* with a if:

- (1) $n^\gamma = n^a$,
- (2) $\forall u \in 2^{n^a} \phi^a(u) \subseteq \phi^\gamma(u)$, and
- (3) $\forall k < n^a \forall v \in 2^{n^a - (k+1)} \psi_k^a(v) \subseteq \psi_k^\gamma(v)$.

We say that a is *Y -terminal* if no σ -configuration is compatible with Y and a one-step extension of a . Given a set $Y \subseteq \kappa^\mathbb{N} \times X$, let $A(a, Y)$ denote the set of pairs of the form $(\sigma, (\phi_\sigma \circ \phi^\gamma)(s_{n^a}))$, where $\sigma \in \kappa^\mathbb{N}$ and γ is a σ -configuration compatible with a and Y_σ .

Lemma 4.1. *Suppose that a is an approximation, $\sigma \in \kappa^\mathbb{N}$, $Y \subseteq \kappa^\mathbb{N} \times X$, and a is Y_σ -terminal. Then $A(a, Y)_\sigma$ is G_σ -independent.*

Proof. See, for example, the proof of [dRM21, Lemma 1.5]. □

Lemma 4.2. *If a is an approximation and $B \subseteq \kappa^\mathbb{N} \times X$ has uniformly Borel sections, then $A(a, B)$ has uniformly analytic sections.*

Proof. The set of pairs (σ, γ) for which γ is a σ -configuration compatible with a is closed, and therefore has uniformly Borel sections by Proposition 1.4, so the set of such pairs for which γ is also compatible with B_σ has uniformly Borel sections by Proposition 1.5, and therefore uniformly analytic sections by Proposition 1.6. The desired result therefore follows from Proposition 1.2. □

Given a set $B \subseteq \kappa^\mathbb{N} \times X$ and $s \in \kappa^{<\mathbb{N}}$, we say that a is *B -terminal on $\mathcal{N}_s$* if $\forall \sigma \in \mathcal{N}_s$ a is B_σ -terminal. Set $A_s(a, B) = A(a, B) \cap (\mathcal{N}_s \times X)$. If B has uniformly Borel sections and a is B -terminal on $\mathcal{N}_s$, then Proposition 3.4 and Lemmas 4.1 and 4.2 yield a G -independent set $A_s(a, B) \subseteq B_s(a, B) \subseteq \mathcal{N}_s \times X$ with uniformly Borel sections.

Let $\mathcal{A}$ denote the set of all approximations. We will recursively construct an increasing sequence $(S^\alpha)_{\alpha < \kappa^+}$ of subsets of $\kappa^{<\mathbb{N}} \times \mathcal{A}$, starting with $S^0 = \emptyset$ and setting $S^\delta = \bigcup_{\alpha < \delta} S^\alpha$ for all limit ordinals $\delta < \kappa^+$. Given $\alpha < \kappa^+$ and $(S^\beta)_{\beta \leq \alpha}$, define $\rho^\alpha: S^\alpha \rightarrow \alpha$ by $\rho^\alpha(s, a) = \min\{\beta < \alpha \mid (s, a) \in S^{\beta+1}\}$, $B^\alpha = \sim \nabla_{s \in \kappa^{<\mathbb{N}}} \bigcup_{(s, a) \in S^\alpha} B_s(a, B^{\rho^\alpha(s, a)})$, and $S^{\alpha+1} = S^\alpha \cup \{(s, a) \in \sim S^\alpha \mid a \text{ is } B^\alpha\text{-terminal on } \mathcal{N}_s\}$, and fix $B_s(a, B^\alpha)$ as above for all $(s, a) \in S^{\alpha+1} \setminus S^\alpha$. As $|\kappa^{<\mathbb{N}} \times \mathcal{A}| = \kappa$, there is a least $\lambda < \kappa^+$ with $S^\lambda = S^{\lambda+1}$. Set $B = B^\lambda$, $S = S^\lambda$, and $\rho = \rho^{\lambda+1}$.

Lemma 4.3. *Suppose that $(s, a) \in \sim S$. Then there exist $\sigma \in \mathcal{N}_s$, a one-step extension b of a , and a σ -configuration γ compatible with b and B_σ such that $(s \frown (\sigma(n^a)), b) \in \sim S$.*

Proof. Fix $\sigma \in \mathcal{N}_s$ for which a is not B_σ -terminal. Then there is a σ -configuration γ compatible with a one-step extension b of a and B_σ .

Suppose, towards a contradiction, that $(s \smallfrown (\sigma(n^a)), b) \in S$. Setting $\beta = \rho(s \smallfrown (\sigma(n^a)), b)$, it follows that b is B^β -terminal on $\mathcal{N}_{s \smallfrown (\sigma(n^a))}$, so $A_{s \smallfrown (\sigma(n^a))}(b, B^\beta) \subseteq B_{s \smallfrown (\sigma(n^a))}(b, B^\beta)$. Then γ is compatible with B_σ^β , so $(\sigma, (\phi_\sigma \circ \phi^\gamma)(s_{n^b})) \in A_{s \smallfrown (\sigma(n^a))}(b, B^\beta) \subseteq B_{s \smallfrown (\sigma(n^a))}(b, B^\beta)$, contradicting the fact that $(\sigma, (\phi_\sigma \circ \phi^\gamma)(s_{n^b})) \in B$. $\square$

Let a_0 denote the unique approximation for which $n^{a_0} = 0$.

Lemma 4.4. *If $(\emptyset, a_0) \notin S$, then there are $\sigma \in \kappa^\mathbb{N}$ and a continuous homomorphism from $\mathbb{G}_0$ to G_σ .*

Proof. Recursively construct $s_n \in \kappa^n$ and approximations a_n with $n^{a_n} = n$ and $(s_n, a_n) \notin S$, beginning with $s_0 = \emptyset$ and a_0 . Given s_n and a_n , apply Lemma 4.3 to obtain $\sigma_n \in \mathcal{N}_{s_n}$, a one-step extension a_{n+1} of a_n , and a σ_n -configuration γ_n compatible with a_{n+1} and B_{σ_n} , and set $s_{n+1} = \sigma_n \upharpoonright n+1$. The same lemma ensures that $(s_{n+1}, a_{n+1}) \notin S$.

Set $\sigma = \bigcup_{n \in \mathbb{N}} s_n$. Then $\sigma_n \rightarrow \sigma$. Define $\phi': 2^\mathbb{N} \rightarrow \mathbb{N}^\mathbb{N}$ and $\psi'_k: 2^\mathbb{N} \rightarrow \mathbb{N}^\mathbb{N}$ by $\phi'(c) = \bigcup_{n \in \mathbb{N}} \phi^{a_n}(c \upharpoonright n)$ and $\psi'_k(c) = \bigcup_{n > k} \psi_k^{a_n}(c \upharpoonright n - (k+1))$. For all $c \in 2^\mathbb{N}$, the sequence $(\sigma_n, \phi^{\gamma_n}(c \upharpoonright n+1))$ converges to $(\sigma, \phi'(c))$, so the closedness of C gives $(\sigma, \phi'(c)) \in C$. Similarly, for all $k \in \mathbb{N}$ and $c \in 2^\mathbb{N}$, the closedness of D gives $(\sigma, \psi'_k(c)) \in D$.

Define $\pi: 2^\mathbb{N} \rightarrow X$ by $\pi(c) = (\text{proj}_X \circ \phi_\sigma)(\phi'(c))$. Passing to the limit in the configuration identities yields $(\text{proj}_{X \times X} \circ \psi_\sigma)(\psi'_k(c)) = (\pi(s_k \smallfrown (i) \smallfrown c))_{i < 2}$ for all $k \in \mathbb{N}$ and $c \in 2^\mathbb{N}$. Thus π is a continuous homomorphism from $\mathbb{G}_0$ to G_σ . $\square$

If $(\emptyset, a_0) \notin S$, then Lemma 4.4 yields (2). Otherwise, set $\alpha = \rho(\emptyset, a_0)$. As $A(a_0, B) = B \cap \bigcup_{i < 2} p_i(G)$, $B \subseteq B^\alpha$, and $A(a_0, B^\alpha) \subseteq B_\emptyset(a_0, B^\alpha)$, the definition of B gives $A(a_0, B) = \emptyset$. Thus B is G -independent. For all $a \in \mathcal{A}$ and $n \in \mathbb{N}$, define $S_{a,n} = \{s \in \kappa^n \mid (s, a) \in S\}$ and $B_{a,n} = \nabla_{s \in S_{a,n}} B_s(a, B^{\rho(s,a)})$. Every $B_{a,n}$ is G -independent, since at a fixed parameter at most one member of $S_{a,n}$ contributes. The fact that $\kappa^\mathbb{N} \times X = B \cup \bigcup_{a \in \mathcal{A}, n \in \mathbb{N}} B_{a,n}$ therefore yields (1).

It only remains to note that if both (1) and (2) hold, then there is a Borel $\mathbb{N}$ -coloring of $\mathbb{G}_0$, contradicting [dRM21, Proposition 1.3]. $\square$

5. STRATIFICATIONS AND CATEGORY TRANSFORMS

Given a set $B \subseteq \kappa^\mathbb{N} \times X$ with uniformly Borel sections and $t \in \kappa^{<\mathbb{N}}$, define $B^{*t} = \{x \in X \mid B^x \text{ is comeager in } \mathcal{N}_t\}$.

Proposition 5.1. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a topological space, $B \subseteq \kappa^\mathbb{N} \times X$ and $B_s \subseteq \kappa^\mathbb{N} \times X$ have uniformly Borel sections for all $s \in \kappa^{<\mathbb{N}}$, and $U \subseteq X$ is open. Then:*

- (1) $(\kappa^\mathbb{N} \times U)^{*t} = U$.

- (2) $\sim B^{*t} = \sim \bigcup_{u \sqsupseteq t} B^{*u}$.
 (3) $(\bigvee_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t} = \bigcap_{u \sqsupseteq t} \bigcup_{v \sqsupseteq u} \bigcup_{n \leq |v|} B_{v \upharpoonright n}^{*v}$.

Proof. As (1) and (2) are straightforward, we need only establish (3). Towards this end, note that if $x \in (\bigvee_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t}$, then $(\bigvee_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is comeager in $\mathcal{N}_t$, so $\forall u \sqsupseteq t \exists n \in \mathbb{N} \bigcup_{s \in \kappa^n} \mathcal{N}_s \cap B_s^x$ is non-meager in $\mathcal{N}_u$, thus $\forall u \sqsupseteq t \exists s \in \kappa^{<\mathbb{N}} \mathcal{N}_s \cap B_s^x$ is non-meager in $\mathcal{N}_u$, and it follows that $\forall u \sqsupseteq t \exists v \sqsupseteq u \exists n \leq |v| B_{v \upharpoonright n}^x$ is comeager in $\mathcal{N}_v$, hence $x \in \bigcap_{u \sqsupseteq t} \bigcup_{v \sqsupseteq u} \bigcup_{n \leq |v|} B_{v \upharpoonright n}^{*v}$. And if $x \notin (\bigvee_{s \in \kappa^{<\mathbb{N}}} B_s)^{*t}$, then $(\bigvee_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is not comeager in $\mathcal{N}_t$, so there exists $u \sqsupseteq t$ such that $(\bigvee_{s \in \kappa^{<\mathbb{N}}} B_s)^x$ is meager in $\mathcal{N}_u$, thus $\forall v \sqsupseteq u \forall n \leq |v| B_{v \upharpoonright n}^x$ is meager in $\mathcal{N}_v$, hence $x \notin \bigcup_{v \sqsupseteq u} \bigcup_{n \leq |v|} B_{v \upharpoonright n}^{*v}$. $\square$

A subset of a topological space is κ^+ -Borel if it belongs to the smallest family containing the open sets and closed under complements and unions of cardinality at most κ .

Proposition 5.2. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a topological space, and $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections. Then B^{*t} is κ^+ -Borel.*

Proof. By the obvious induction using Proposition 5.1. $\square$

A set $A \subseteq X$ is κ -Souslin if there exist a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ with the property that $A = \phi(C)$. We say that a set $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a *stratification* of a κ -Souslin set A if there are witnesses $C \subseteq \kappa^{\mathbb{N}}$ and $\phi: C \rightarrow X$ to the κ -Souslinity of A with the property that $A' = \phi'(C')$, where $C' = \{(\sigma, b) \in \kappa^{\mathbb{N}} \times \mathbb{N}^{\mathbb{N}} \mid \sigma \circ b \in C\}$ and $\phi': C' \rightarrow \kappa^{\mathbb{N}} \times X$ is given by $\phi'(\sigma, b) = (\sigma, \phi(\sigma \circ b))$. For each $\sigma \in \kappa^{\mathbb{N}}$, we view A'_σ as the *analytic trace* of A on the set $\sigma(\mathbb{N})$.

Proposition 5.3. *Suppose that κ is an aleph, X is a Hausdorff space, $A \subseteq X$ is κ -Souslin, $A' \subseteq \kappa^{\mathbb{N}} \times X$ is a stratification of A , and $x \in A$. Then $\forall^* \sigma \in \kappa^{\mathbb{N}} x \in A'_\sigma$.*

Proof. Fix a closed set $C \subseteq \kappa^{\mathbb{N}}$ and a continuous function $\phi: C \rightarrow X$ with the property that $A = \phi(C)$ and $A' = \phi'(C')$, as well as $\tau \in C$ for which $\phi(\tau) = x$. Then the set $H = \bigcap_{n \in \mathbb{N}} \bigcup_{s \in \kappa^{<\mathbb{N}}} \mathcal{N}_{s \frown (\tau \upharpoonright n)}$ is a dense G_δ subset of $\kappa^{\mathbb{N}}$ and $\forall \sigma \in H x \in A'_\sigma$. $\square$

Proposition 5.4. *Suppose that κ is an aleph, $t \in \kappa^{<\mathbb{N}}$, X is a Hausdorff space, G is a κ -Souslin digraph on X , G' is a stratification of G , $B \subseteq \kappa^{\mathbb{N}} \times X$ has uniformly Borel sections, and B_σ is G'_σ -independent for every σ . Then B^{*t} is G -independent.*

Proof. Otherwise fix $x, y \in B^{*t}$ with $x G y$ and appeal to Proposition 5.3 to obtain $\sigma \in \mathcal{N}_t$ with $x, y \in B_\sigma$ and $x G'_\sigma y$, a contradiction. $\square$

Theorem 5.5 (Kanovei). *Suppose that κ is an infinite cardinal, X is a Hausdorff space, and G is a κ -Souslin digraph on X . Then at least one of the following holds:*

- (1) *There is a κ^+ -Borel coloring $c: X \rightarrow \kappa$ of G .*
- (2) *There is a continuous homomorphism from $\mathbb{G}_0$ to G .*

Proof. Fix a stratification G' of G . As every homomorphism from $\mathbb{G}_0$ to a vertical section of G' is a homomorphism from $\mathbb{G}_0$ to G , Theorem 1 allows us to assume there is a weakly Borel function $c': \kappa^\mathbb{N} \times X \rightarrow \mathbb{N}$ such that $\forall \sigma \in \kappa^\mathbb{N}$ c'_σ is a coloring of G'_σ . For all $n \in \mathbb{N}$, set $B_n = (c')^{-1}(\{n\})$. For all $t \in \kappa^{<\mathbb{N}}$, Proposition 5.2 ensures that B_n^{*t} is κ^+ -Borel and Proposition 5.4 ensures that it is G -independent. It only remains to note that if $x \in X$, then $\kappa^\mathbb{N} = \bigcup_{n \in \mathbb{N}} B_n^x$, so there exists $n \in \mathbb{N}$ for which B_n^x is non-meager. Proposition 2.1 and localization therefore yield $t \in \kappa^{<\mathbb{N}}$ for which B_n^x is comeager in $\mathcal{N}_t$, thus $x \in B_n^{*t}$. $\square$

As noted in [Mil12, §8], the weakening of Theorem 5.5 without the definability condition on c follows from the simplification of the classical proof of the $\mathbb{G}_0$ dichotomy in which the use of separation is removed. As separation easily generalizes to κ -Souslin sets, this suggests a simpler route towards a classical proof of the full version of Theorem 5.5. That route, however, would require the generalization of separation to continuous images of κ^+ -Borel sets. To see that this is consistently false, note first that every Π_2^1 set is $\aleph_2$ -Borel (see, for example, [Kec95, Exercise 38.8]), so every Σ_3^1 set is a continuous image of an $\aleph_2$ -Borel set. By [HS85, Claim 8], we can assume that $\text{add}(\text{meager}) \geq \aleph_2$ and there is a Δ_3^1 set B without the Baire property. But then every $\aleph_2$ -Borel set has the Baire property, so B and $\sim B$ are disjoint continuous images of $\aleph_2$ -Borel sets that cannot be separated by an $\aleph_2$ -Borel set.

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